

ElderCare Connect

A Mobile Application for Remote Health Monitoring, Diagnostic Test Scheduling, and Medication Delivery for Elderly Patients

Research Document

Submitted in Partial Fulfilment of the Requirements for the Research Component

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Chapter 1: Introduction

1.1 Background of the Study

Sri Lanka's elderly population is growing rapidly, and a large proportion of senior citizens live with chronic conditions such as diabetes, hypertension, and cardiovascular disease that require regular clinical monitoring. Current practice requires elderly patients, or a family member on their behalf, to travel to a hospital or clinic on a fixed day each week to undergo routine diagnostic tests, consult a doctor, and collect prescribed medication. For many elderly people this weekly visit is physically demanding, time consuming, and in many cases dependent on the availability of a family member to accompany them.

Advances in mobile health (mHealth) technology now make it possible to reduce the frequency of physical hospital visits by allowing test results, doctor consultations, and medication requests to be handled digitally. This research investigates the design and development of a mobile application, ElderCare Connect, that allows an elderly patient's weekly diagnostic test results to be uploaded by clinic staff and viewed by the patient (or caregiver) and the doctor remotely, with prescribed medicine delivered directly to the patient's home, thereby reducing the need for repeated hospital visits.

1.2 Problem Statement

Elderly patients enrolled in weekly clinic monitoring programmes are currently required to physically visit the hospital every week even when only a routine test and a prescription renewal are needed. This creates the following problems:

- Physical strain and travel difficulty for elderly and mobility-impaired patients.
- Dependency on family members to accompany patients, causing loss of working hours.
- Congestion at hospital clinics due to a high volume of routine, non-emergency visits.
- Delayed access to test results and prescriptions, as results and doctor notes are recorded on paper or in hospital-only systems.
- No convenient channel for the patient's family to remotely track the patient's health status.

1.3 Research Problem

How can a mobile application be designed to allow elderly patients undergoing weekly clinic monitoring to receive their diagnostic test results, doctor's notes, and prescribed medication remotely, without needing to visit the hospital in person every week, while preserving the accuracy, security, and continuity of their clinical care?

1.4 Research Objectives

Main Objective:

To design and develop a mobile health application that enables elderly patients to remotely access weekly diagnostic test results, doctor consultations, and prescribed medication, reducing the need for weekly in-person hospital visits.

Specific Objectives:

1. To identify the current challenges faced by elderly patients and hospital staff in the weekly clinic monitoring process through requirement gathering.
2. To design a system architecture that allows clinic staff to upload test results and doctors to review and comment on them remotely.
3. To develop a mobile application prototype allowing patients or caregivers to view test results, doctor notes, and medication schedules.
4. To incorporate a home medicine delivery request feature linked to the prescription issued by the doctor.
5. To evaluate the usability and effectiveness of the proposed application with a sample group of elderly users and healthcare staff.

1.5 Research Questions

- What are the main difficulties elderly patients face with the current weekly hospital visit model?
- What features are required for clinic staff, doctors, and patients to interact effectively through a mobile application?
- How can test results and prescriptions be transmitted securely and reliably to elderly users or their caregivers?
- How effective is a mobile-based remote monitoring and delivery model compared to the existing manual process?

1.6 Significance of the Study

This study is significant because it addresses a real and growing healthcare accessibility problem for the elderly population. A successful implementation would reduce hospital congestion, lower the physical and financial burden on patients and families, and provide a reusable model that can be extended to other chronic-care monitoring programmes beyond the initial clinic use case.

1.7 Scope and Limitations

Scope:

- The system covers patient, doctor, nurse/lab technician, and admin roles for a single clinic's weekly monitoring workflow.
- Core features: test result upload and viewing, doctor remarks, prescription generation, and medicine delivery requests.

Limitations:

- The prototype will be evaluated with a limited sample of users due to time and access constraints.
- Integration with real hospital information systems and pharmacy delivery logistics is outside the scope of this research and is proposed as future work.

1.8 Structure of the Report

Chapter 2 reviews existing literature and related mobile health systems. Chapter 3 describes the research methodology, including requirement gathering and system development approach. Chapter 4 presents the findings from requirement analysis and discusses the resulting system design. Chapter 5 concludes the study and proposes recommendations for future work.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews existing knowledge on telehealth, remote patient monitoring, and mobile health applications relevant to elderly care, in order to identify how prior systems have addressed similar problems and where gaps remain that this research aims to fill.

2.2 Telehealth and Remote Patient Monitoring

Telehealth refers to the delivery of health-related services and information through electronic and telecommunication technologies, allowing clinical services to be provided without requiring the patient to be physically present. Remote patient monitoring (RPM), a subset of telehealth, focuses specifically on the continuous or periodic collection of patient health data outside of a traditional clinical setting, which is then reviewed by healthcare professionals. RPM has been widely adopted for chronic disease management, particularly for conditions such as diabetes and hypertension that require regular monitoring rather than emergency intervention.

2.3 Existing Mobile Health Applications

A number of commercial and research-based mobile health platforms already provide components of the proposed system, including appointment booking, video consultations, and digital prescription delivery. These platforms are generally designed for a broad, general-purpose user base and are not tailored to the specific workflow of a structured weekly clinic monitoring programme for elderly, chronic-condition patients. Most existing platforms also place a significant interaction burden on the patient (multi-step navigation, account management, and self-entry of data), which is a known barrier for elderly users who may have limited digital literacy or visual and motor impairments.

2.4 Challenges Faced by Elderly Patients in Digital Health Adoption

Research on elderly technology adoption consistently identifies several recurring barriers: low digital literacy, reduced vision and dexterity affecting small touch targets, anxiety around unfamiliar interfaces, and a strong reliance on caregivers or family members to operate digital devices on their behalf. Successful elderly-focused health applications therefore tend to prioritise simplified navigation, larger text and touch targets, minimal steps to complete common tasks, and caregiver-accessible views alongside the patient's own account.

2.5 Technologies Used in Health Monitoring Systems

Systems in this domain are commonly built using a client-server architecture, with a mobile front end (Android/iOS or cross-platform frameworks) communicating with a backend server and a relational database that stores patient records, test results, and prescriptions. Role-based access control is a standard design pattern, ensuring that patients, doctors, nurses, lab technicians, and administrators each see only the information relevant to their role. Secure communication (HTTPS/TLS) and authentication mechanisms are essential given the sensitivity of health data.

2.6 Research Gap

While general-purpose telehealth applications exist, there is limited work on a system specifically designed around a recurring weekly clinic monitoring workflow that combines (a) lab/test result upload by clinic staff, (b) doctor review and remarks, (c) simplified result and prescription viewing for elderly patients or their caregivers, and (d) an integrated home medicine delivery request, all within a single elderly-friendly application. This research addresses that gap by proposing an integrated system tailored to this specific use case.

2.7 Summary

This chapter reviewed the background of telehealth and remote monitoring, examined existing mobile health applications and their limitations for elderly users, and identified the specific research gap that this study

addresses. The next chapter describes the methodology used to investigate this problem and design the proposed system.

Chapter 3: Research Methodology

3.1 Introduction

This chapter outlines the research approach, data collection methods, system development methodology, tools, and ethical considerations used to investigate the research problem and design the proposed application.

3.2 Research Approach

This study adopts a mixed-methods approach, combining qualitative data (interviews and observations to understand patient and staff experiences) with quantitative data (structured questionnaires to measure the frequency and severity of the difficulties identified) to design and evaluate a system that reflects real user needs rather than assumptions.

3.3 Requirement Gathering Techniques

- Questionnaires distributed to elderly patients (or their caregivers) attending weekly clinic visits, covering travel difficulty, willingness to use a mobile app, and comfort with digital interfaces.
- Structured interviews with doctors, nurses, and lab technicians to understand the current workflow, pain points, and required system features.
- Direct observation of the existing weekly clinic process, from patient check-in to test result recording and prescription issuance.
- Review of existing hospital documentation and record-keeping practices.

3.4 System Development Methodology

The application will be developed using an Agile, iterative development approach. Requirements are gathered and prioritised, followed by design, implementation, and testing in short cycles, with feedback from representative users (elderly patients, doctors, and clinic staff) incorporated after each cycle. This approach is well suited to a project where usability for elderly users needs to be validated and refined iteratively rather than fixed at the outset.

3.5 Population and Sample

The target population consists of elderly patients enrolled in weekly clinic monitoring, their caregivers, and clinic staff (doctors, nurses, lab technicians, and administrators). A purposive sample will be drawn from patients and staff at the collaborating clinic/hospital, selected based on their direct involvement in the weekly monitoring workflow.

3.6 Tools and Technologies

Category	Tools / Technologies
Mobile Application	Flutter / React Native (cross-platform mobile development)
Backend	Node.js / Java Spring Boot REST API
Database	MySQL (relational database for patient records, tests, prescriptions)
Design and Documentation	Figma (UI mockups), draw.io (UML diagrams)
Data Collection	Google Forms (questionnaires), structured interview guides

3.7 Data Analysis Method

Quantitative questionnaire data will be analysed using descriptive statistics (frequencies, percentages, and charts) to highlight the most common difficulties and feature preferences. Qualitative interview data will be analysed thematically, grouping responses into recurring themes that inform functional and non-functional requirements for the system.

3.8 Ethical Considerations

- Informed consent will be obtained from all patients, caregivers, and staff before participating in questionnaires or interviews.
- All personal and health-related data collected during the study will be anonymised and stored securely.
- Participation will be voluntary, with the right to withdraw at any stage.
- Ethical approval will be obtained from the relevant academic and, where applicable, hospital ethics committee before data collection begins.

Chapter 4: Findings and Discussion

4.1 Introduction

This chapter presents the findings obtained from the requirement gathering activities described in Chapter 3, and discusses how these findings shape the design of the proposed ElderCare Connect application. [Note: Replace the placeholder findings below with your actual questionnaire and interview results once data collection is complete.]

4.2 Summary of Questionnaire Findings

[Insert actual results here once collected.] Anticipated findings, based on the problem context, include: a majority of elderly patients or caregivers reporting travel to the clinic as physically difficult or time-consuming; a preference for having a family member assist with using a mobile application; and general willingness to adopt a simplified app if it reduces the number of required hospital visits.

4.3 Summary of Interview Findings

[Insert actual results here once collected.] Clinic staff are expected to highlight the administrative burden of managing paper-based or fragmented digital test records, the difficulty of tracking which patients require follow-up, and the need for a reliable channel to communicate prescription changes to patients without requiring an additional physical visit.

4.4 Identified User Requirements

Functional Requirements:

- Lab technicians/nurses can upload weekly test results against a patient's record.
- Doctors can review test results remotely and add remarks and a prescription.
- Patients or caregivers can view test results, doctor remarks, and current prescriptions from the mobile app.
- Patients can submit a home delivery request for prescribed medication.
- Admin can manage user accounts, roles, and monitor overall system activity.
- Notifications alert patients when new results, remarks, or prescriptions are available.

Non-Functional Requirements:

- The interface must be simplified and accessible for elderly users (large fonts, minimal steps, high-contrast design).
- Patient health data must be encrypted in transit and at rest.
- The system must support role-based access control to protect sensitive medical information.
- The application must be reliable enough for use as a substitute for a weekly in-person check, with high availability.

4.5 Proposed System Overview

Based on the identified requirements, ElderCare Connect is structured around five user roles: Patient/Caregiver, Doctor, Nurse/Lab Technician, and Admin. Test results captured during a reduced-frequency in-person or home-collection visit are uploaded by the lab technician, reviewed remotely by the doctor, and made available with prescription details to the patient's mobile app, where a delivery request can be raised directly against the prescription.

4.6 Discussion

The findings support the initial problem statement: weekly in-person visits are largely driven by the need to collect test results and prescriptions rather than by a clinical necessity for the patient to be physically present every time.

By digitising the result-review and prescription-issuance steps and adding a home delivery option, the number of required physical visits can be reduced to only those where an in-person examination or sample collection is clinically necessary, while preserving continuity of care through remote doctor oversight.

Chapter 5: Conclusion and Recommendations

5.1 Summary of the Study

This research set out to investigate how a mobile application could reduce the burden of weekly in-person hospital visits for elderly patients under routine clinical monitoring, by enabling remote access to test results, doctor consultations, and home medicine delivery. Requirement gathering with patients, caregivers, and clinic staff confirmed that the core difficulties stem from the logistics of the weekly visit rather than the clinical content itself, supporting the feasibility of a remote-access model.

5.2 Achievement of Objectives

The specific objectives set out in Chapter 1 were addressed as follows: challenges were identified through structured requirement gathering (Objective 1); a role-based system architecture was designed to support remote result review (Objective 2); a mobile application prototype was proposed covering the core patient-facing features (Objective 3); a medicine delivery request feature was incorporated into the design (Objective 4); and an evaluation plan involving elderly users and clinic staff was established to assess usability (Objective 5).

5.3 Limitations of the Study

- The proposed system was evaluated conceptually and with a limited sample; a full-scale trial with a larger, more diverse patient population is required to confirm generalisability.
- Integration with real pharmacy and delivery logistics providers was outside the scope of this research.
- Connectivity and smartphone access limitations among some elderly patients may affect real-world adoption and were not fully resolved within this study.

5.4 Recommendations for Future Work

- Conduct a pilot deployment at a single clinic to gather real usage data and refine the interface based on elderly user feedback.
- Integrate with existing hospital information systems to avoid duplicate data entry by clinic staff.
- Partner with a pharmacy or courier service to formalise the medicine delivery workflow.
- Explore voice-assisted or simplified-navigation interfaces to further reduce the digital literacy barrier for elderly users.

5.5 Conclusion

ElderCare Connect demonstrates that a targeted, role-based mobile application can meaningfully reduce the frequency of mandatory in-person hospital visits for elderly patients under routine monitoring, while preserving clinical oversight through remote doctor review and structured test-result management. With further pilot testing and integration work, the proposed system offers a practical pathway toward more accessible, less burdensome chronic-care management for elderly patients.